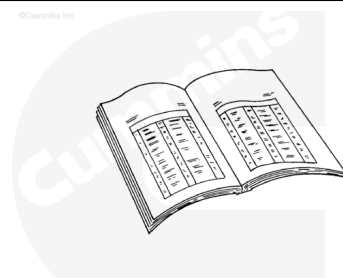


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>)
- Remove the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>)
- Remove the rocker levers. Refer to Procedure 003-008 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>)
- Remove the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/100/100-004-014.html>)
- Remove the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>)
- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001->

031.html)

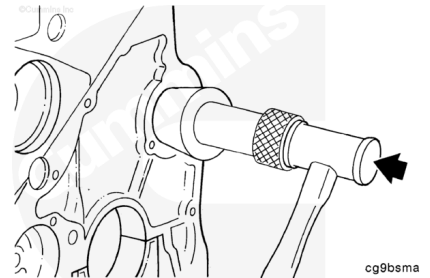
- Remove the camshaft. Refer to Procedure 001-008 in Section 1. (/qs3/pubsys2/xml/en/procedures/41/41-001-008.html)

Remove

Drive the camshaft bushing from the cylinder block bore with camshaft bushing removal and installation tool, Part Number 3165045.

The following kits are also available for camshaft bushing installation and removal:

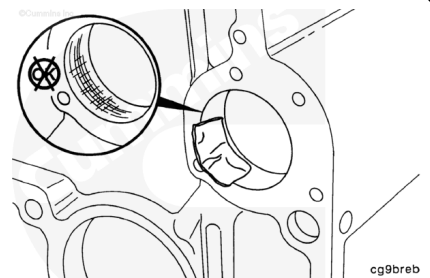
- Hydraulic Actuator Kit, Part Number 3823621
- Camshaft Bushing Kit, Part Number 3162253



Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



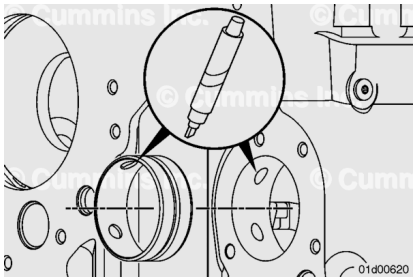
Use a fine emery cloth and solvent to remove any burrs and sharp edges from the camshaft bushing bores.

Install

Mark the camshaft bushing and cylinder block to aid in the alignment of align the lubricating oil hole in the cylinder block with the lubricating oil hole in the bushing.

Note : The top hole on the front and rear camshaft bushings will not line up with the top hole of the

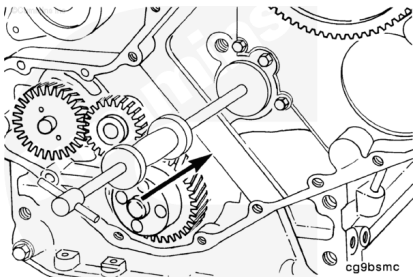
cylinder block bores. Only the bottom hole will be line up for the front and rear camshaft bushings.



Use the camshaft bushing installation and removal tool, Part Number 3165045, to install the camshaft bushing so that it is even with the front face of the cylinder block.

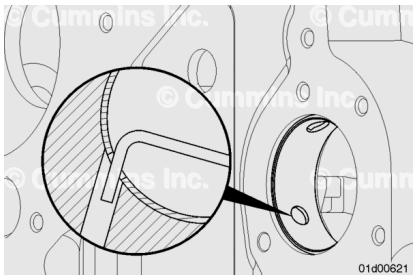
For ease of bushing installation use Hydraulic Actuator Kit, Part Number 3823621, and Camshaft Bushing Kit, Part Number 3162253.

Note : The front camshaft bushing **must** be installed to a depth of 4.5 mm [0.177 in] below the front face of the cylinder block. All other bushings **must** be installed even with the web of the camshaft bores.



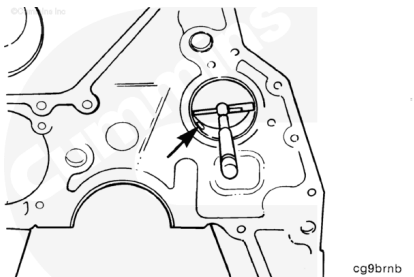
For the front and rear camshaft bushings, **only** the bottom hole will line up with the hole in the cylinder block. Make sure the lubricating oil hole is aligned.

A 3.2 mm [0.128 in] diameter rod **must** be able to pass freely through the camshaft bushing oil hole.



Measure the installed diameter of the camshaft bushings.

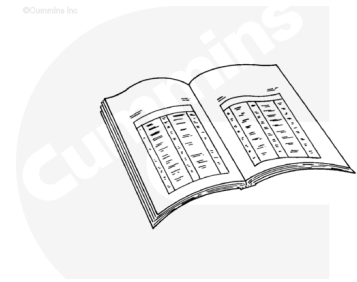
Camshaft Bushing Bore Diameter - Installed		
mm		in
60.06	MIN	2.365
60.12	MAX	2.367



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the camshaft. Refer to Procedure 001-008 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/41/41-001-031.html>)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-052.html>)
- Install the push rods. Refer to Procedure 004-014 in Section 4. (</qs3/pubsys2/xml/en/procedures/100/100-004-014.html>)
- Install the rocker levers. Refer to Procedure 003-008 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>)
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-002.html>)
- Connect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Operate the engine and check for leaks.

